

A PROJECT REPORT ON

Design and Implementation of a UART Based
Password Security System on FPGA

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Osmania University, Hyderabad-500007

Submitted by

V. Vivek Saradhi

A. Raj Kumar

S. Anand Babu

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Table of Contents

ABSTRACT	5
CHAPTER 1	6
INTRODUCTION	6
1.1 Introduction ..	6
1.2 Overview of FPGA	6
1.3 Introduction to UART Communication	7
1.4 Need for Password Security Systems	7
1.5 Problem Statement	8
1.6 Objectives of the Project	8
1.7 Scope of the Project	8
CHAPTER 2	10
LITERATURE SURVEY	10
2.1 Introduction	10
2.2 Password Security Systems	10
2.3 Traditional Password Security Systems	11
2.4 FPGA Technology	11
2.5 FPGA-Based Password Security Systems	12
2.6 UART Communication	12
2.7 Finite State Machine (FSM) Based Design	13
2.8 Password Update Mechanism	14
2.9 Counter and Timer Mechanisms	15
2.10 UART Communication Process.....	16
2.11 Comparison of FPGA and Microcontroller Approaches	16
CHAPTER 3	17
SYSTEM ANALYSIS	17
3.1 Introduction	17
3.2 Existing System	17
3.3 Limitations of Existing System	17
3.4 Need for Proposed System	18
3.5 Proposed System	18
3.6 Features of Proposed System	19
3.7 System Requirements	19
3.8 Functional Requirements	20

3.9 State Analysis	20 3.10
Advantages of Proposed System	21
CHAPTER 4	22
SYSTEM DESIGN	22
4.1 Introduction	22
4.2 Overall System Architecture	22
4.3 Block Diagram	23
4.4 Description of System Blocks	23
4.5 Flow Diagram Description.....	24
4.6 Finite State Machine Design	26
4.7 FSM State Diagram	27
4.8 Failed attempt detection and lock mechanism	27
4.9 Timing Analysis	28
CHAPTER 5	29
MODULE DESCRIPTION	29
5.1 Introduction	29
5.2 Main Module	29
5.3 UART receiver Module (uart receiver)	29
5.4 RX Controller Module (rx_controller)	30
CHAPTER 6	31
IMPLEMENTATION AND TESTING	31
6.1 Introduction	31
6.2 Software Used	31
6.3 Design Entry	31
6.4 Simulation	32
6.5 Working of the System	32
6.6 Synthesis	33
6.7 Implementation	33
6.8 Bitstream Generation	34
6.9 Hardware Testing	34
6.10 Test Cases	35
6.11 Results	36
CHAPTER 7	37
RESULTS AND DISCUSSION	37

7.1 Introduction	37
7.2 Simulation Results	37
7.3 Hardware Results	38
7.4 Password Authentication Result	38
7.5 Password update Result	38
7.6 Discussion	40
7.7 Advantages of the Developed System	40
7.8 Applications	40
CHAPTER 8	40
CONCLUSION AND FUTURE SCOPE	41
Conclusion	41
8.2 Future Scope	41
8.3 Final Remarks	42
References	43

ABSTRACT

The increasing demand for secure access control systems has led to the development of efficient and reliable password-based security solutions. This project, titled "Design and Implementation of a UART Based Password Security System on FPGA," presents a digital security system implemented using Verilog HDL on an FPGA platform.

The system allows users to enter a password through UART communication using a serial terminal. The entered password is compared with a stored password, and access is granted only when the correct password is entered. To enhance security, the system maintains a count of consecutive failed login attempts. After three incorrect attempts, the system enters a lock state and activates a buzzer for a predefined duration, preventing unauthorized access.

An additional feature of the system is password updating through UART communication. Authorized users can modify the existing password after successful verification of the current password. Status messages such as SUCCESS, FAIL, and LOCK are transmitted back to the user through UART, providing real-time feedback.

The complete design was modeled using Verilog HDL, simulated, synthesized, and successfully implemented on FPGA hardware. The project demonstrates the practical application of Finite State Machines (FSMs), UART communication, digital counters, timers, and hardware-based security mechanisms in modern digital systems.

CHAPTER 1

INTRODUCTION

1.1 Introduction

Security systems are widely used to protect information, devices, and restricted areas from unauthorized access. Password-based authentication is one of the most common and reliable methods used in modern security systems.

Field Programmable Gate Arrays (FPGAs) provide a flexible platform for implementing digital designs with high speed and reliability. UART (Universal Asynchronous Receiver Transmitter) is a simple serial communication protocol used for data exchange between electronic devices.

This project presents the design and implementation of a UART-Based Password Security System on FPGA using Verilog HDL. The system allows users to enter a password through a serial terminal and verifies it against a stored password. To improve security, the system locks after three consecutive incorrect attempts and activates a buzzer for a predefined period.

The system also supports password updating through UART communication after successful verification of the existing password. Status messages such as SUCCESS, FAIL, and LOCK are transmitted to the user through UART. The project demonstrates the practical implementation of UART communication, Finite State Machines (FSMs), counters, timers, and digital security concepts on FPGA hardware.

1.2 Overview of FPGA

A Field Programmable Gate Array (FPGA) is an integrated circuit that can be programmed and reconfigured after manufacturing to perform specific digital functions. It consists of configurable logic blocks, programmable interconnections, and input/output resources that allow designers to implement complex digital systems

FPGA technology offers:

- High processing speed
- Parallel execution
- Real-time operation
- Reconfigurable architecture
- Reliable hardware implementation

FPGAs are widely used in communication systems, embedded systems, industrial In this project, the FPGA is used to implement a UART-based Password Security System.

It provides reliable operation, fast password verification, lock control, timer functionality, and UART communication through hardware implementation.automation, aerospace applications, and digital signal processing.

1.3 Introduction to UART Communication

Universal Asynchronous Receiver Transmitter (UART) is a widely used serial communication protocol that enables data transfer between digital devices. UART communication is asynchronous, meaning that it does not require a separate clock signal between the transmitter and receiver. Instead, both devices operate at a predefined baud rate.

UART communication uses two primary signals:

- TX (Transmit)
- RX (Receive)

Data is transmitted serially, one bit at a time, making UART a simple and efficient communication method for embedded systems, microcontrollers, and FPGA-based designs.

In this project, UART communication is used to receive password data from a serial terminal and transmit status messages such as SUCCESS, FAIL, and LOCK back to the user. This enables easy interaction between the FPGA system and a computer terminal without requiring additional hardware interfaces.

The use of UART simplifies user communication and provides a reliable method for password entry, verification, and password update operations in the proposed security system.

1.4 Need for Password Security Systems

Password security systems are widely used to protect devices, information, and restricted areas from unauthorized access. They provide a simple and reliable method of authentication.

Home and Office Security

Used to restrict access to authorized users and improve security.

Industrial Applications

Used to protect equipment, control systems, and confidential data.

Educational Applications

Provide practical understanding of FPGA design, UART communication, FSMs, counters, and timers.

Importance of FPGA Implementation

FPGA-based security systems offer high speed, reliability, and hardware-level security.

The proposed UART-Based Password Security System demonstrates an effective and secure authentication system implemented on FPGA.

1.5 Problem Statement

Traditional password security systems may be vulnerable to unauthorized access and often lack reliable mechanisms for handling repeated incorrect password attempts. There is a need for a hardware-based security system capable of:

- Secure password authentication
- Real-time password verification
- Detection of multiple incorrect attempts
- Automatic lock mechanism after repeated failures
- Password update functionalit

The proposed FPGA-based UART Password Security System addresses these requirements through dedicated hardware implementation, providing reliable and secure access control.

1.6 Objectives of the Project

The main objectives of the project are:

- To design a UART-based Password Security System using FPGA.
- To implement a Finite State Machine (FSM) using Verilog HDL.
- To verify the entered password with a stored password.
- To provide a lock mechanism after three consecutive incorrect attempts.
- To implement password update functionality through UART communication.
- To verify the design through simulation and FPGA implementation.

1.7 Scope of the Project

The project focuses on implementing a password authentication system using FPGA technology.

The scope includes:

- Verilog HDL design
- UART communication
- FSM implementation
- Password verification
- Counter and timer design
- Password update feature
- FPGA hardware implementation

- Simulation verification

The project does not include wireless communication, data storage, or advanced graphical interfaces. However, these features can be added in future enhancements.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

A literature survey is an important phase of project development that provides an understanding of existing technologies, methodologies, and implementation techniques related to the proposed system. The purpose of this chapter is to study password security systems, FPGA technology, UART communication, and authentication mechanisms used in similar applications.

Password-based security systems are widely used in homes, offices, industries, and embedded systems to restrict unauthorized access. Various implementations have been developed using microcontrollers, processors, and FPGA platforms. FPGA-based implementations offer advantages such as high speed, parallel processing, reliability, and reconfigurability.

The proposed UART-Based Password Security System combines password authentication, UART communication, Finite State Machine (FSM) control, counters, and timers to provide a secure and reliable access control solution implemented on FPGA hardware.

2.2 Password Security Systems

Password security systems are widely used to restrict unauthorized access to devices, information, and restricted areas. These systems authenticate users by comparing an entered password with a stored password.

A typical password security system consists of:

- Password input unit
- Password verification unit
- Access control unit
- User feedback unit

The basic operation involves receiving a password from the user, verifying it against the stored password, and granting or denying access based on the result.

Several password security systems have been developed using:

- Mechanical locking systems
- Microcontroller-based systems
- Embedded processors
- FPGA-based digital systems

To improve security, many modern systems include features such as multiple login attempt detection, automatic lock mechanisms, and password update functionality.

The reliability and security of the system depend on the authentication mechanism and the hardware platform used for implementation.

2.3 Traditional Password Security Systems

Traditional password security systems are commonly implemented using microcontrollers such as:

- Arduino
- PIC Microcontrollers
- AVR Controllers
- ARM-Based Controllers

These systems receive user passwords, verify them through software, and provide access control based on authentication results.

Advantages

- Easy implementation
- Low cost
- Simple programming

Disadvantages

- Software execution delay
- Limited parallel processing
- Lower reliability compared to hardware implementation
- Reduced security due to software dependency

Since password verification requires reliable and fast operation, software overhead can affect system performance. FPGA-based implementations overcome these limitations by providing dedicated hardware for authentication and access control.

2.4 FPGA Technology

Field Programmable Gate Arrays (FPGAs) are integrated circuits that can be configured after manufacturing to perform specific digital functions.

Unlike microcontrollers that execute instructions sequentially, FPGAs implement hardware directly, allowing multiple operations to occur simultaneously.

An FPGA consists of:

- Configurable Logic Blocks (CLBs)
- Input/Output Blocks (IOBs)
- Programmable Interconnects
- Clock Management Units

The behavior of the FPGA is defined using Hardware Description Languages such as:

- Verilog HDL

- VHDL

FPGA technology provides significant advantages in timing-sensitive applications such as reaction timer systems.

2.5 FPGA-Based Password Security Systems

Many researchers and designers implement password security systems using FPGA technology due to its advantages such as:

- High processing speed
- Parallel execution
- Hardware-level implementation
- Reliable operation

FPGA-based password security systems generally use:

- Finite State Machines (FSMs)
- UART communication modules
- Digital counters and timers
- Password verification circuits

The authentication process involves receiving a password, verifying it against a stored password, and granting or denying access based on the result.

Compared to software-based systems, FPGA implementations provide better performance, reliability, and security through dedicated hardware implementation.

2.6 UART Communication

Universal Asynchronous Receiver Transmitter (UART) is a widely used serial communication protocol for data exchange between digital devices. UART communication does not require a separate clock signal and operates using predefined baud rates.

A UART communication system consists of:

- UART Transmitter (TX)
- UART Receiver (RX)

The communication process involves:

1. Sending data serially through the transmitter.
2. Receiving data through the receiver.

3. Converting serial data into parallel form for processing.

4. Transmitting status messages back to the user.

In the proposed system, UART communication is used to receive passwords from a serial terminal and transmit status messages such as SUCCESS, FAIL, and LOCK. This provides a simple and reliable interface between the FPGA and the user.

2.7 Finite State Machine (FSM) Based Design

Finite State Machines are commonly used in digital systems to control sequential operations.

FSMs consist of:

- States
- Inputs
- Outputs
- State transitions

The proposed project uses FSM architecture to control:

- Password verification
- Login success and failure detection
- Lock mechanism after multiple failed attempts
- Password update operations
- Message generation

The FSM implemented in the project contains the following states:

- IDLE
- INPUT
- VERIFY
- SUCCESS
- FAIL
- LOCK
- UPDATE_VERIFY
- UPDATE_PASSWORD

Each state performs a specific function and transitions to the next state based on user inputs and system conditions.

FSM-based design improves:

- System reliability
 - Modular design
 - Easy debugging
 - Efficient hardware implementation
-

2.8 Password Update Mechanism

Password update functionality enhances the flexibility and usability of a security system by allowing authorized users to change the existing password without modifying the hardware design.

The password update process implemented in the proposed system consists of:

- Update request generation through UART
- Verification of the existing password
- Entry of a new password
- Updating the stored password
- Confirmation of successful update

This feature improves system security by allowing periodic password changes and preventing unauthorized access through old credentials.

The password update mechanism is implemented using UART communication, FSM control, and password verification logic, making the system more practical for real-world security applications.

2.9 Counter and Timer Mechanisms

Counters and timers are important components in digital security systems. They are used to monitor events and control time-dependent operations.

In the proposed UART-Based Password Security System, a counter is used to track consecutive incorrect password attempts. If the number of failed attempts reaches three, the system enters a lock state to prevent unauthorized access.

A timer is used to control the lock duration. Once the lock period expires, the system automatically returns to normal operation and allows the user to enter a password again.

The use of counters and timers improves system security and reliability by preventing repeated unauthorized login attempts and providing controlled access recovery.

2.10 UART Communication Process

UART (Universal Asynchronous Receiver Transmitter) is a serial communication protocol used for data exchange between digital devices.

The communication process involves:

1. Transmitting password data through the UART transmitter.
2. Receiving data through the UART receiver.
3. Converting serial data into parallel form.
4. Processing the received password.
5. Transmitting status messages back to the user.

UART communication is simple, reliable, and widely used in embedded and FPGA-based systems. In the proposed project, UART serves as the interface between the FPGA and the serial terminal for password entry and status display.

2.11 Comparison of FPGA and Microcontroller Approaches

Feature	FPGA	Microcontroller
Processing	Parallel	Sequential
Speed	High	Moderate
Real-Time Performance	Excellent	Good
Hardware Control	Direct	Software Based
Reliability	High	Moderate
Reconfigurability	High	Limited

The FPGA approach is well suited for password security applications because authentication, counters, timers, and UART communication are implemented directly in hardware, resulting in fast and reliable operation.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Introduction

System analysis is the process of studying existing systems to identify their limitations and develop an improved solution. In this project, a UART-Based Password Security System is developed using FPGA technology to provide secure and reliable password authentication.

The objective of this analysis is to understand the limitations of traditional password security systems and justify the need for an FPGA-based implementation.

3.2 Existing System

Traditional password security systems are commonly implemented using microcontrollers, processors, or software-based applications. These systems generally authenticate users through software routines and provide access based on password verification.

A typical password security system consists of:

- Microcontroller unit
- Password input interface
- Verification software
- User feedback mechanism

Although these systems are widely used, they suffer from certain limitations related to software execution and reliability.

3.3 Limitations of Existing System

1. Software Execution Delay

Microcontroller systems execute instructions sequentially. The timing measurement may be affected by program execution delays and interrupt handling.

2. Limited Processing Speed

The response time depends on processor clock frequency and software efficiency.

3. Timing Inaccuracy

Software-based systems are more susceptible to unauthorized modifications and security breaches compared to dedicated hardware implementations.

4. Limited Parallel Processing

Microcontrollers can perform only a limited number of tasks simultaneously, whereas FPGA systems execute multiple operations in parallel.

5. Reduced Reliability

System performance may decrease when multiple software tasks execute simultaneously, affecting authentication and access control operations.

3.4 Need for Proposed System

The limitations of traditional password security systems create the need for a hardware-based authentication system that provides:

- Secure password verification
- Fast response time
- Real-time operation
- Reliable access control
- Password update functionality
- Enhanced system security

The proposed FPGA-based UART Password Security System overcomes these limitations by implementing password authentication, counters, timers, and UART communication directly in hardware.

3.5 Proposed System

The limitations of traditional password security systems create the need for a hardware-based authentication system that provides:

- Secure password verification
- Fast response time
- Real-time operation
- Reliable access control
- Password update functionality
- Enhanced system security

The proposed FPGA-based UART Password Security System overcomes these limitations by implementing password authentication, counters, timers, and UART communication directly in hardware.

The user must immediately press the stop button. The measured reaction time is stored and displayed on the seven-segment display.

If the user presses the stop button before the timer officially starts, the system detects a false start and displays an error value.

3.6 Features of Proposed System

The limitations of traditional password security systems create the need for a hardware-based authentication system that provides:

- Secure password verification
- Fast response time
- Real-time operation
- Reliable access control
- Password update functionality
- Enhanced system security

The proposed FPGA-based UART Password Security System overcomes these limitations by implementing password authentication, counters, timers, and UART communication directly in hardware.

3.7 System Requirements

Hardware Requirements

The hardware components required for the project are:

Component	Purpose
FPGA Development Board	Main controller and implementation platform
USB-UART Interface	Serial communication between PC and FPGA
Buzzer	Indicates lock condition
USB cable	Programming the FPGA board
USB-UART Interface	Serial communication between the PC terminal

Software Requirements

The software tools required are:

Software	Purpose
Xilinx Vivado	Designandimplementation
Verilog HDL	Hardwaredescription
Vivado Simulator	Simulationand verification

3.8 Functional Requirements

The system must satisfy the following functional requirements:

Password Entry Function

The system shall accept a password from the user through UART communication.

password Verification Function

The entered password shall be compared with the stored password to determine whether access should be granted or denied.

Authentication Function

The system shall generate SUCCESS or FAIL messages based on the verification result.

Failed Attempt Monitoring Function

The system shall count consecutive incorrect password attempts.

Lock Function

The system shall enter a lock state and activate the buzzer after three consecutive incorrect password attempts.

Password Update Function

The system shall allow authorized users to update the existing password after successful verification.

Reset Function

The system shall return to its initial state when the reset signal is activated.

3.9 State Analysis

The proposed UART-Based Password Security System uses a Finite State Machine (FSM) consisting of the following states:

IDLE

The system enters the initial state after reset and waits for user interaction.

INPUT

The system waits for the user to enter a password through UART communication.

VERIFY

The entered password is compared with the stored password.

SUCCESS

This state is entered when the correct password is received. A SUCCESS message is transmitted to the user.

FAIL

This state is entered when the password verification fails. A FAIL message is transmitted to the user.

LOCK

The system enters this state after three consecutive incorrect password attempts. The buzzer is activated and access is temporarily blocked.

UPDATE_VERIFY

The system verifies the current password before allowing a password update operation.

UPDATE_PASSWORD

The system accepts and stores a new password after successful verification of the existing password.

The FSM ensures organized system operation and controls password verification, lock functionality, UART message generation, and password update operations.

3.10 Advantages of Proposed System

Compared with traditional password security systems, the proposed FPGA-based implementation offers:

- High-speed operation
- Parallel processing capability
- Hardware-level authentication
- Improved security
- Reliable performance
- Real-time password verification
- Password update functionality
- Efficient access control

These advantages make FPGA technology highly suitable for implementing secure and reliable password authentication systems.

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design is the process of defining the architecture, modules, interfaces, and data flow of a system to satisfy the specified requirements. The design phase converts the project requirements into a practical hardware implementation.

The proposed UART-Based Password Security System is designed using FPGA technology and Verilog HDL. The complete system is controlled using a Finite State Machine (FSM) that manages password verification, authentication, lock control, password update operations, and UART communication.

The design consists of multiple interconnected modules such as UART Receiver, RX Controller, Comparator, FSM Controller, Counter, Timer, Message Controller, and UART Transmitter. These modules work together to provide secure password authentication and reliable access control. The complete design was implemented, simulated, and verified on FPGA hardware to ensure correct system operation.

4.2 Overall System Architecture

The proposed UART-Based Password Security System consists of the following major blocks:

UART Receiver
RX Controller
Comparator
FSM Controller
Counter
Timer
Message Controller
UART Transmitter
Buzzer

The FPGA acts as the central controller that coordinates all system operations.

System Operation Sequence

1. The user enters a password through a serial terminal.
2. The UART Receiver receives the serial data.
3. The RX Controller converts the received data into a password format.
4. The Comparator compares the entered password with the stored password.
5. The FSM Controller verifies the authentication result.
6. If the password is correct, a SUCCESS message is transmitted.
7. If the password is incorrect, a FAIL message is transmitted and the failed-attempt counter is incremented.

8. After three consecutive incorrect attempts, the system enters the LOCK state and activates the Buzzer.
9. The timer controls the lock duration and restores normal operation after timeout.
10. Authorized users can update the password through UART communication.

4.3 Block Diagram

Block Diagram of Proposed System

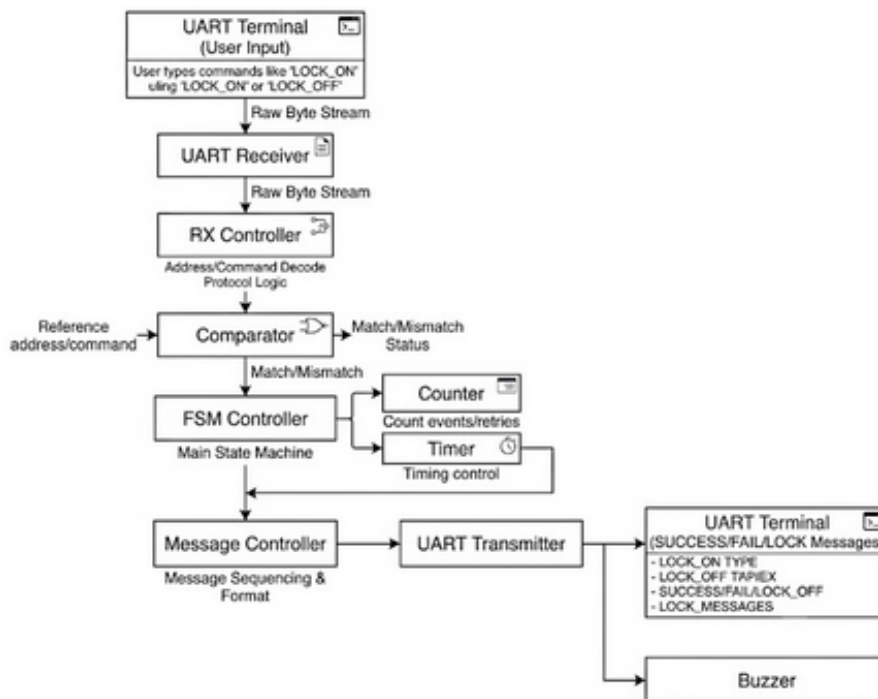


fig-4.1

4.4 Description of System Blocks

1. UART Receiver

The UART Receiver receives serial data from the terminal and converts it into parallel data. It generates a ready signal after successfully receiving a character.

2. RX Controller

The RX Controller processes the received UART data and forms the password. It also generates control signals such as password_valid and update_request.

3. Comparator

The Comparator compares the entered password with the stored password and generates a match signal when both passwords are equal.

4. FSM Controller

The Finite State Machine (FSM) acts as the central control unit of the system. It controls password verification, authentication, failed-attempt monitoring, lock operation, and password update functionality.

5. Counter

The Counter keeps track of consecutive incorrect password attempts. When the count reaches three, it activates the lock mechanism.

6. Timer

The Timer controls the lock duration. After the timer expires, the system returns to normal operation.

7. Message Controller

The Message Controller generates status messages such as SUCCESS, FAIL, and LOCK based on the current system state.

8. UART Transmitter

The UART Transmitter sends status messages from the FPGA to the serial terminal for user feedback.

9. Buzzer

The buzzer is activated during the lock state to indicate that access has been temporarily blocked due to multiple incorrect password attempts.

4.3 Flowchart Description

The flowchart illustrates the overall operation of the UART-Based Password Security System. The process begins when the system is powered on or reset. The FPGA enters a waiting state and waits for the user to enter a password through UART communication.

The received password is processed by the UART Receiver and RX Controller and then compared with the stored password using the Comparator module. If the password matches, the system generates a SUCCESS message and grants access. If the password does not match, a FAIL message is generated and the failed-attempt counter is incremented.

When the number of consecutive failed attempts reaches three, the system enters the LOCK state. In this state, the buzzer is activated and a timer is started. The system remains locked until the timer expires, after which it returns to the password entry state.

The system also supports password updating. An authorized user can initiate the update operation through a UART command. After successful verification of the current password, the user can enter a new password, which is stored in place of the old password. A confirmation message is then transmitted to indicate successful password update.

The flowchart demonstrates the complete authentication, lock control, and password update operations implemented in the FPGA-based security system.

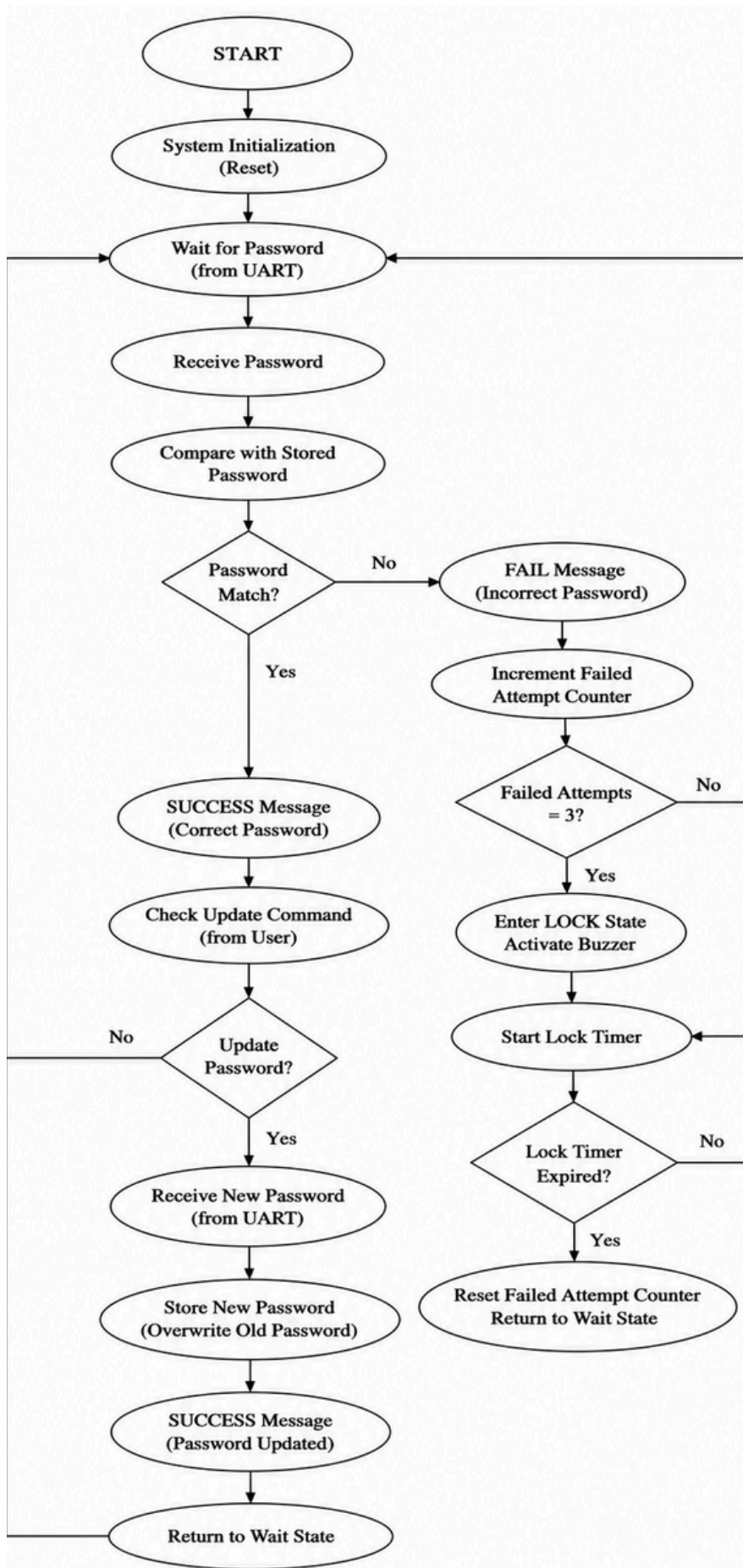


fig-4.2

4.6 Finite State Machine (FSM) Design

The proposed UART-Based Password Security System is implemented using a Finite State Machine (FSM). The FSM acts as the central controller and manages all system operations.

The FSM controls:

- Password verification
- Failed attempt monitoring
- Lock mechanism
- Password update operation
- UART message generation

FSM States

IDLE

Initial state after system reset.

INPUT

Waits for the user to enter a password through UART.

VERIFY

Compares the entered password with the stored password.

SUCCESS

Indicates successful authentication and generates a SUCCESS message.

FAIL

Indicates incorrect password entry and generates a FAIL message.

LOCK

Activated after three consecutive failed attempts. The buzzer is enabled and a timer is started.

UPDATE_VERIFY

Verifies the current password before allowing password modification.

UPDATE_PASSWORD

Accepts and stores the new password after successful verification.

The FSM ensures proper state transitions and coordinates all authentication, security, and communication operations within the system.

4.7 FSM State Diagram

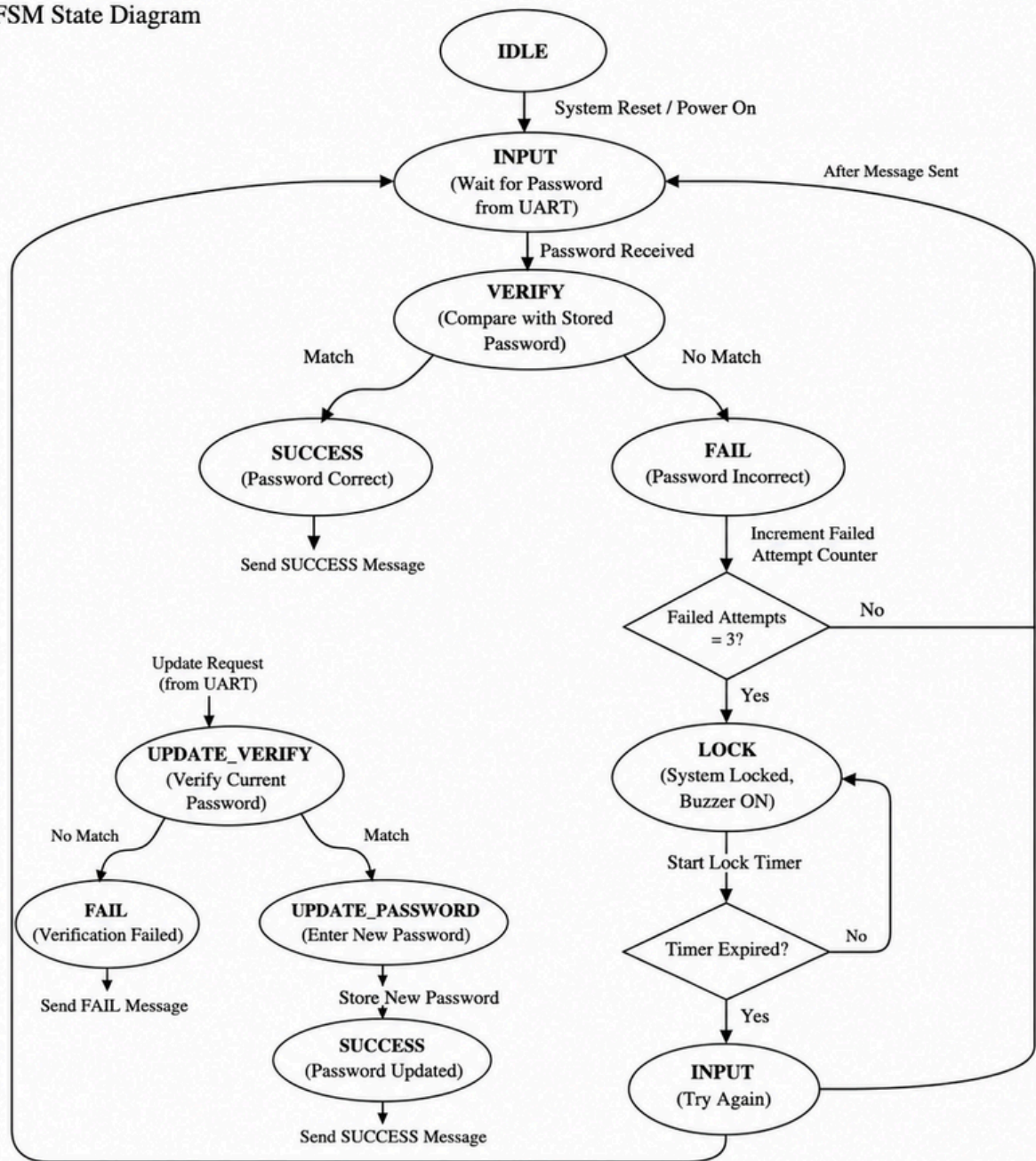


fig-4.3

4.8 Failed Attempt Detection and Lock Mechanism

To improve security, the system monitors consecutive incorrect password attempts. Whenever the entered password does not match the stored password, the failed-attempt counter is incremented.

If the number of consecutive failed attempts reaches three, the system enters the LOCK state. In this state, the buzzer is activated and a timer is started. The user is prevented from further authentication attempts until the lock timer expires.

The lock mechanism is implemented using a counter and timer controlled by the Finite State Machine (FSM). This feature helps prevent unauthorized access through repeated password guessing attempts.

The functionality can be represented as:

```
if(!match)
increment <= 1'b1;
if(attempts_done)
next_state <= lock;
```

The counter is cleared after a successful password verification, ensuring that only consecutive failed attempts are considered.

4.9 Timing Analysis

The proposed UART-Based Password Security System operates using the onboard FPGA clock. The timing performance of the system depends on UART communication, password verification, failed-attempt counting, and lock timer operation.

FPGA Clock

Clock Frequency = 100 MHz

Clock Period = 10 n sec

$$T = 1/f = 1/100 * 10^6 = 10 * 10^{-9} = 10 \text{ n sec}$$

Password Verification Timing

After receiving a complete password through UART communication, the Comparator verifies the entered password with the stored password within a few clock cycles, providing fast authentication.

Failed Attempt Detection

Whenever an incorrect password is entered, the failed-attempt counter is incremented. After three consecutive incorrect attempts, the FSM transitions to the LOCK state.

Lock Timer Operation

The lock timer is activated when the system enters the LOCK state. During this period, further authentication attempts are blocked. After the timer expires, the FSM returns to the INPUT state and allows password entry again.

UART Communication Timing

The UART Transmitter and UART Receiver operate according to the configured baud rate. Status messages such as SUCCESS, FAIL, and LOCK are transmitted serially to the terminal.

The timing analysis confirms that the system provides reliable authentication, secure lock operation, and efficient UART communication using FPGA hardware implementation.

CHAPTER 5

MODULE DESCRIPTION

5.1 Introduction

The UART-Based Password Security System is implemented in Verilog HDL and consists of several modules for password authentication and UART communication. The main modules include the Main Module, UART Receiver, RX Controller, Comparator, FSM Controller, Counter, Timer, Message Controller, and UART Transmitter. Together, these modules receive user passwords, verify authentication, generate status messages, manage lock functionality, and support password updates. The design follows a modular architecture, making it easier to test, debug, and enhance individual components without affecting the overall system operation. Furthermore, the system is designed to provide secure access control by monitoring failed login attempts and activating protective measures such as temporary system lock and buzzer alerts when unauthorized access is detected.

5.2 Main Module

Purpose

This is the top-level module of the project. It controls the complete operation of the UART-Based Password Security System and integrates all submodules.

Functions

- Interfaces all system modules.
- Stores the current password.
- Controls password update operations.
- Generates message trigger signals.
- Manages authentication flow.
- Controls communication between modules.

Working

Initially, the system stores a default password. The user enters a password through the UART terminal. The received password is verified against the stored password using the comparator module.

If the password is correct, the system generates a SUCCESS message. If the password is incorrect, a FAIL message is generated and the failed-attempt counter is incremented. After three consecutive failed attempts, the system enters the LOCK state and activates the buzzer for a predefined duration.

The main module also supports password updating. An authorized user can verify the current password and replace it with a new password through UART communication.

5.3 UART Receiver Module (uart_receiver)

Purpose

This module receives serial data from the UART terminal and converts it into parallel data for further processing.

Functions

- Receives serial data through the RX line.
- Detects start and stop bits.
- Converts serial data into parallel format.
- Generates a ready signal when data reception is complete.
- Transfers received data to the RX Controller.

Working

The UART Receiver continuously monitors the RX line for incoming serial data. When a valid start bit is detected, the module samples the incoming bits according to the configured baud rate and reconstructs the transmitted byte.

After receiving all data bits and the stop bit, the module stores the received character and generates a ready signal. The received data is then forwarded to the RX Controller for password processing.

5.4 RX Controller Module (rx_controller)

Purpose

This module processes the data received from the UART Receiver and converts it into a password format suitable for verification.

Functions

- Receives data from the UART Receiver.
- Collects password digits entered by the user.
- Generates a password valid signal.
- Detects password update requests.
- Transfers the completed password to the Comparator module.

Working

The RX Controller monitors the data received from the UART Receiver. When valid password digits are received, the module stores them sequentially and forms the complete password.

After receiving all password digits, the module generates a password valid signal indicating that the password is ready for verification. The module also detects special commands used for password update operations and informs the main control logic accordingly.

CHAPTER 6

IMPLEMENTATION AND TESTING

6.1 Introduction

Implementation involves converting the system design into hardware using Verilog HDL and FPGA technology. The UART-Based Password Security System was designed, simulated, synthesized, implemented, and tested on an FPGA board.

The implementation process includes:

- Verilog HDL coding
- Functional simulation
- Synthesis
- Implementation
- Bitstream generation
- FPGA programming
- Hardware testing
- The design was developed in Xilinx Vivado and verified through simulation before FPGA deployment. Hardware testing validated password authentication, lock control, password updates, and UART communication.

6.2 Software Used

The following software tools were used in this project:

Software	Purpose
Xilinx Vivado	Design and Implementation
Verilog HDL	Hardware Description Language

Xilinx Vivado

Xilinx Vivado was used to write the Verilog HDL code, perform simulation, synthesize the design, generate the bitstream file, and program the FPGA board.

Verilog HDL

Verilog HDL was used to design and implement the UART-Based Password Security System modules, including UART communication, FSM control, password verification, counters, timers, and message generation.

Vivado Simulator

Vivado Simulator was used to verify the functionality of the design through waveform analysis before hardware implementation.

6.3 Design Entry

The project was developed using Verilog HDL.

The following files were created:

- main.v
- uart.v
- uart_receiver.v
- rx_controller.v
- comparator.v
- fsm.v
- counter.v
- timer.v
- message_controller.v

These files were added to the Xilinx Vivado project and verified for syntax correctness before simulation and implementation.

Each module was designed and tested individually before being integrated into the complete UART-Based Password Security System.

6.4 Simulation

Simulation was performed to verify the functionality of the design before hardware implementation.

Simulation Steps

- Create a Vivado project.
- Add all Verilog source files.
- Add the testbench file.
- Run behavioral simulation.
- Observe waveform outputs.
- Verify password authentication, lock functionality, and password update operations.

The simulation confirmed that the system correctly performs password verification, generates SUCCESS and FAIL messages, activates the LOCK state after three consecutive incorrect attempts, and supports password update functionality.

6.5 Working of the System

The operation of the system is as follows:

Step 1

The system initializes and enters the IDLE state after reset.

Step 2

The user enters a password through the UART terminal.

Step 3

The UART Receiver receives the serial data and forwards it to the RX Controller.

Step 4

The RX Controller forms the password and generates a password valid signal.

Step 5

The Comparator compares the entered password with the stored password.

Step 6

If the password is correct, the FSM enters the SUCCESS state and a SUCCESS message is transmitted through UART.

Step 7

If the password is incorrect, the FSM enters the FAIL state and a FAIL message is transmitted.

Step 8

The Counter tracks consecutive incorrect password attempts.

Step 9

After three consecutive incorrect attempts, the FSM enters the LOCK state and activates the buzzer for a predefined duration.

Step 10

An authorized user can update the password through UART communication. The system first verifies the current password and then stores the new password.

Step 11

After successful authentication or completion of the lock period, the system returns to normal operation and waits for the next password entry.

6.6 Synthesis

After successful simulation, the design was synthesized using Xilinx Vivado.

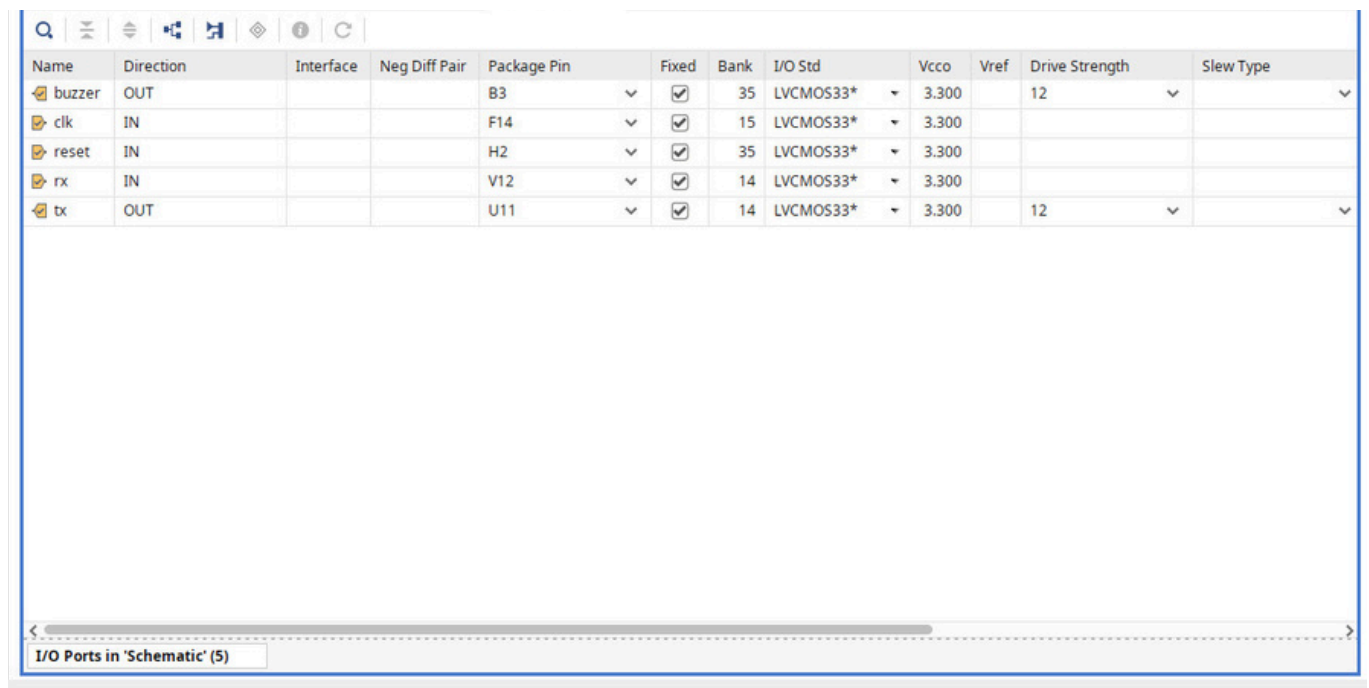
Synthesis converts the Verilog HDL code into FPGA hardware components such as:

- Logic Gates
- Flip-Flops
- Counters
- Multiplexers
- Finite State Machine Logic
- UART Communication Logic

The synthesis report confirmed that the design was implemented successfully without errors.

6.7 Implementation

Pin Configuration:



Name	Direction	Interface	Neg Diff Pair	Package Pin	Fixed	Bank	I/O Std	Vcco	Vref	Drive Strength	Slew Type
buzzer	OUT			B3	☒	35	LVC MOS33*	3.300		12	
clk	IN			F14	☒	15	LVC MOS33*	3.300			
reset	IN			H2	☒	35	LVC MOS33*	3.300			
rx	IN			V12	☒	14	LVC MOS33*	3.300			
tx	OUT			U11	☒	14	LVC MOS33*	3.300		12	

fig-6.1

I/O Port Assignment

The figure shows the Vivado I/O Ports Window after synthesis. It displays the mapping of input and output signals of the UART-Based Password Security System to the FPGA package pins.

Input Signal Assignments

Signal	Direction	Function
rst	Input	System clock input
clk	Input	Resets the system
rx	Input	UART receive signal

6.7 I/O Port Assignment

The figure shows the Vivado I/O Ports window after synthesis. It displays the mapping of the input and output signals of the UART-Based Password Security System to the FPGA package pins.

Input Signal Assignments

Signal Direction Function

clk	Input	System clock input
reset	Input	Resets the system
rx	Input	UART receive signal

Output Signal Assignments

Signal Direction Function

tx	Output	UART transmit signal
buzzer	Output	Indicates lock condition after multiple failed password attempts

The Vivado synthesis tool successfully mapped all input and output ports to the corresponding FPGA package pins. The clock signal drives the entire system, while the UART receive (rx) and transmit (tx) signals enable serial communication between the FPGA and the terminal. The buzzer output is activated during the lock state to indicate unauthorized access attempts.

6.8 Bitstream Generation

After successful synthesis and implementation, the bitstream file was generated using Xilinx Vivado.

The bitstream file contains the configuration data required to program the FPGA device. It represents the final hardware implementation of the UART-Based Password Security System.

The generated bitstream was downloaded to the FPGA board using Vivado Hardware Manager. After programming, the system was tested using UART communication to verify password authentication, lock functionality, password update operations, and message transmission.

6.9 Hardware Testing

The programmed FPGA board was tested using:

- FPGA Development Board
- UART Terminal
- USB-UART Communication
- Buzzer Output

Observations

- Passwords were received correctly through UART.
- Correct passwords generated the SUCCESS message.
- Incorrect passwords generated the FAIL message.
- The failed-attempt counter operated correctly.
- The system entered the LOCK state after three consecutive incorrect attempts.
- The buzzer was activated during the lock period.
- Password update functionality operated correctly.
- UART message transmission worked successfully.

The hardware results matched the simulation results.

6.10 Test Cases

Test Case	Expected Result	Status
Enter Correct Password	SUCCESS message displayed	Passed
Enter Incorrect Password	FAIL message displayed	Passed
Enter Three Consecutive Incorrect Passwords	LOCK state activated	Passed
Buzzer Activation During Lock State	Buzzer turns ON	Passed
Lock Timer Expiration	System returns to normal operation	Passed
Password Update Request	Update mode activated	Passed
Correct Old Password Entry	Password update allowed	Passed
Incorrect Old Password Entry	Password update denied	Passed
New Password Entry	Password updated successfully	Passed
Verify Updated Password	Authentication successful	Passed
Reset System	Returns to initial state	Passed

All test cases were successfully verified through simulation and FPGA hardware testing.

6.11 Results

The UART-Based Password Security System was successfully implemented and tested on the FPGA board.

The system achieved:

- Correct FSM operation
- Successful password authentication
- Reliable UART communication
- Accurate password verification
- Correct generation of SUCCESS and FAIL messages
- Proper failed-attempt counting
- Successful lock activation after three consecutive incorrect attempts
- Correct buzzer operation during lock state
- Successful password update functionality
- Reliable hardware implementation and testing

The simulation and hardware results confirmed the correct operation of all modules and demonstrated the effectiveness of the proposed FPGA-based password security system.

CHAPTER 7

RESULTS AND DISCUSSION

7.1 Introduction

This chapter presents the results obtained from the simulation and hardware implementation of the UART-Based Password Security System. The functionality of the system was verified through simulation and FPGA hardware testing.

The results demonstrate the successful operation of password authentication, UART communication, failed-attempt detection, lock mechanism, buzzer activation, and password update functionality. The system was tested under various conditions, including correct password entry, incorrect password entry, multiple failed login attempts, and password update operations.

The simulation and hardware results confirm that the proposed design operates correctly and provides reliable password-based access control using FPGA technology.

7.2 Simulation Results

The UART-Based Password Security System was simulated using the Vivado Simulator to verify its functionality before hardware implementation.

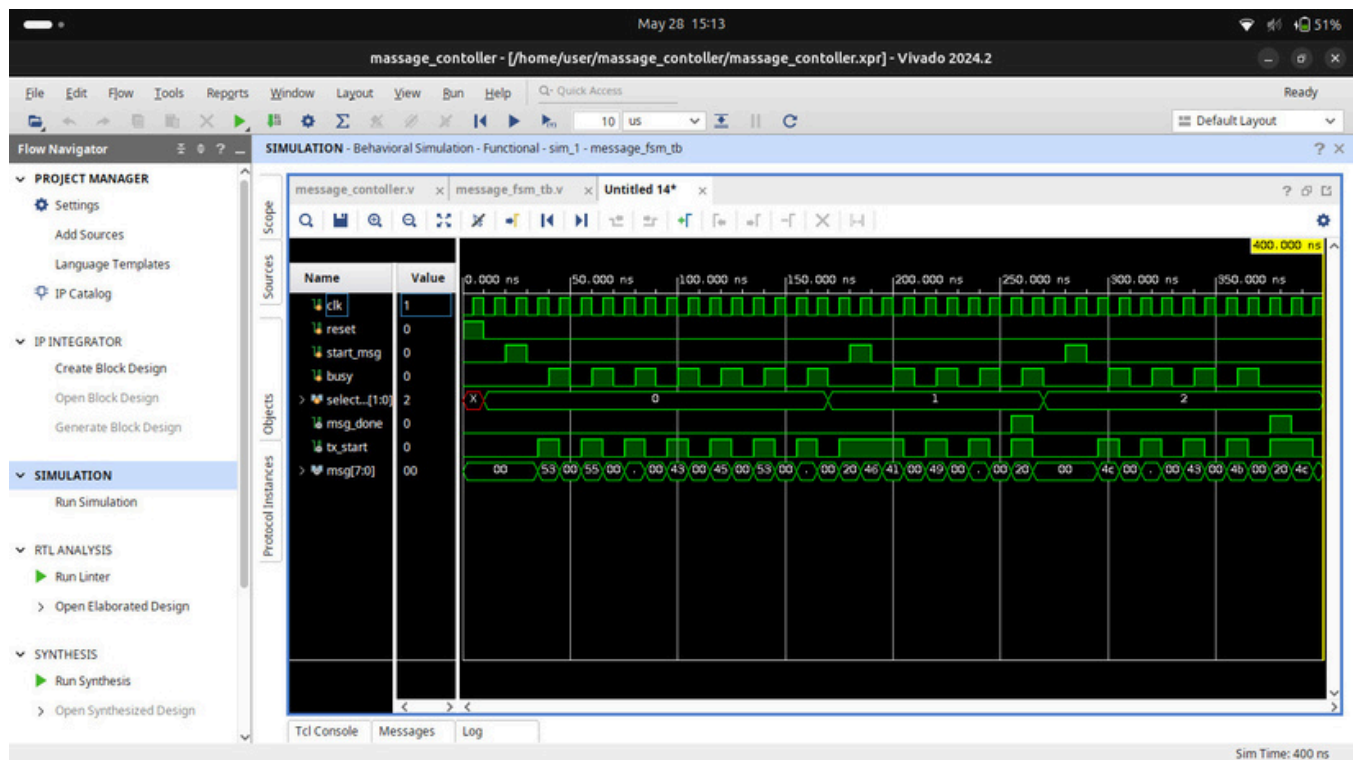


fig-7.1

The simulation confirmed:

- Correct FSM state transitions.
- Successful password verification.
- Proper generation of SUCCESS and FAIL messages.
- Accurate failed-attempt counting.
- Correct activation of the LOCK state after three consecutive incorrect attempts.
- Proper buzzer activation during the lock period.
- Successful password update operation.
- Reliable UART communication.

The waveform results showed that the system operated correctly under different password authentication and password update conditions.

7.3 Hardware Results

After successful simulation, the generated bitstream was programmed into the FPGA board.

The hardware implementation produced the expected results. Passwords were successfully received through UART communication, authentication was performed correctly, and status messages were transmitted back to the user. The lock mechanism and password update functionality were also verified successfully on the FPGA hardware.

7.4 Password Authentication Result

The password authentication functionality was successfully verified on the FPGA board.

Correct Password Verification

When the user entered the correct password through the UART terminal, the system successfully authenticated the user and transmitted a SUCCESS message.

Incorrect Password Verification

When an incorrect password was entered, the system denied access and transmitted a FAIL message.

Multiple Failed Attempts

The failed-attempt counter correctly tracked consecutive incorrect password entries. After three consecutive failed attempts, the system entered the LOCK state and activated the buzzer.

The hardware results confirmed the correct operation of the authentication and security mechanisms.

7.5 Password Update Result

The password update functionality was successfully tested on the FPGA board.

The update process consisted of:

- Entering the update command.
- Verifying the current password.
- Entering a new password.
- Storing the new password.

After the update operation, the system successfully authenticated users using the newly stored password while rejecting the previous password.

The results confirmed that the password update mechanism operated correctly and reliably.

UART Terminal Output Result

The UART terminal successfully displayed the status messages generated by the FPGA-based Password Security System.

The terminal displayed:

- SUCCESS message for correct password entry.
- FAIL message for incorrect password entry.
- LOCK message after three consecutive incorrect password attempts.
- Password update confirmation after successful password modification.

PC Terminal Output:

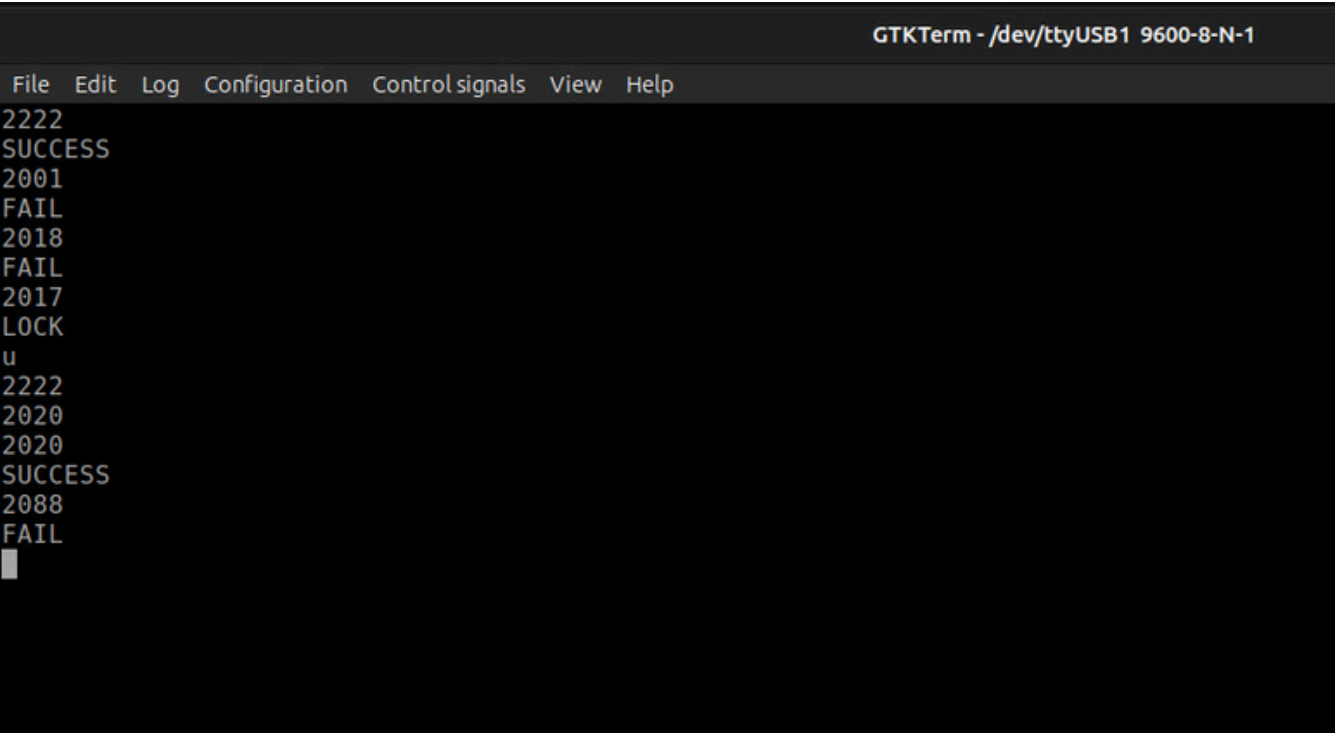


fig-7.2

Lock State Buzzer Output

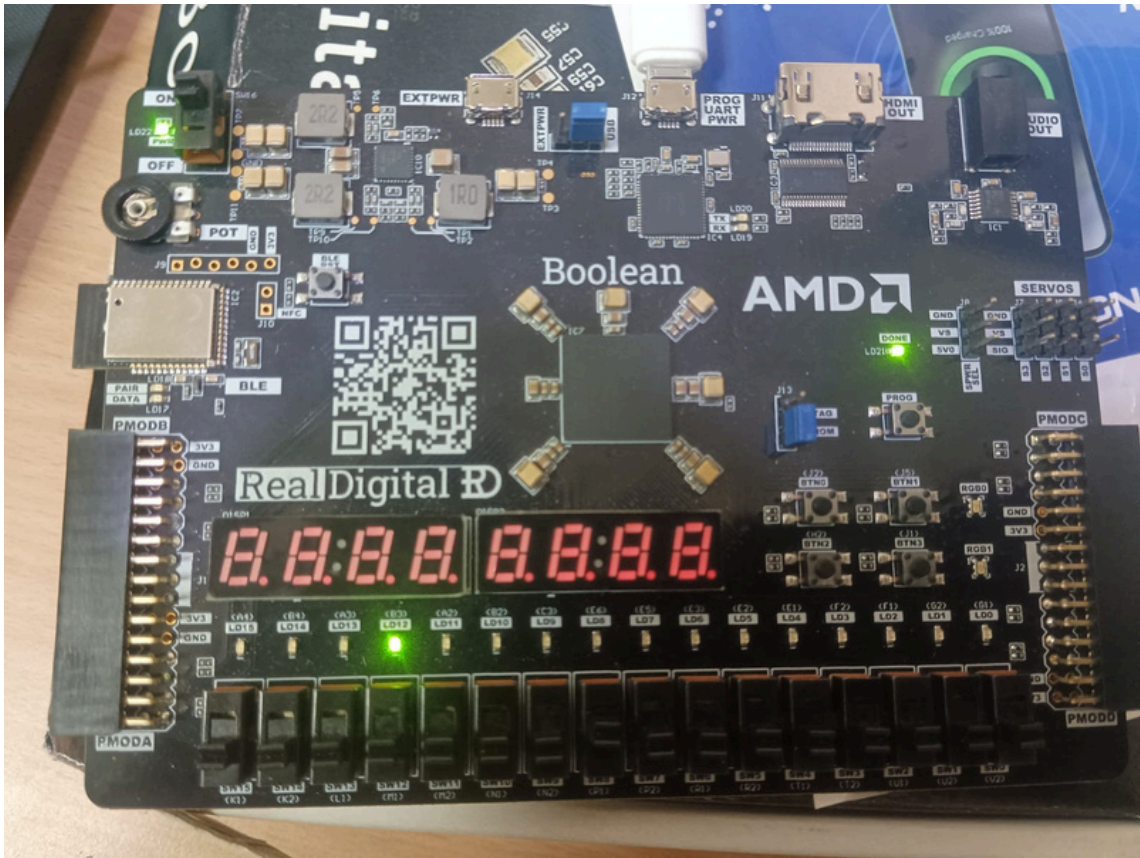


fig-7.3

7.6 Discussion

The proposed UART-Based Password Security System successfully achieved all design objectives. The simulation and hardware results confirmed the correct operation of all modules.

Observations

- The FSM controlled the system effectively.
- Password verification operated correctly.
- UART communication functioned reliably.
- SUCCESS and FAIL messages were generated correctly.
- The failed-attempt counter operated as expected.
- The LOCK mechanism activated after three consecutive incorrect attempts.
- The buzzer operated correctly during the lock period.
- Password update functionality worked successfully.
- The FPGA implementation matched the simulation results.

The use of FPGA technology provided reliable and efficient hardware-based password authentication.

7.7 Advantages of the Developed System

The developed system offers several advantages:

- Secure password authentication.
- Fast hardware-based operation.
- Reliable UART communication.
- Password update capability.
- Automatic lock mechanism.
- Improved system security.
- Real-time operation.
- Modular FPGA design.

7.8 Applications

The developed system can be used in:

- Electronic door lock systems.
- Access control systems.
- Industrial security systems.
- Embedded security applications.
- Educational FPGA projects.
- Laboratory access control.
- Password-protected digital systems.
- Secure authentication systems.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion:

The UART-Based Password Security System was successfully designed and implemented using Verilog HDL on an FPGA platform. The project was developed to provide secure password authentication, access control, and password update functionality through UART communication.

The system receives passwords through a serial terminal, verifies them against a stored password, and generates appropriate status messages such as SUCCESS, FAIL, and LOCK. A lock mechanism was implemented to enhance security by blocking access after three consecutive incorrect password attempts. The system also supports password updating after successful verification of the existing password.

The project successfully achieved all its objectives:

Password authentication was implemented successfully.

UART communication operated correctly.

The Finite State Machine (FSM) controlled the complete system effectively.

Failed-attempt detection and lock functionality worked as expected.

Password update functionality was implemented successfully.

Simulation and FPGA hardware results matched the expected behavior.

The FPGA-based implementation provided reliable, secure, and real-time operation. The project also enhanced understanding of Verilog HDL, FSM design, UART communication, counters, timers, simulation, and FPGA implementation techniques.

Overall, the developed system serves as an effective password-based security solution and demonstrates the practical application of digital design concepts using FPGA technology.

8.2 Future Scope

The proposed UART-Based Password Security System can be further enhanced by incorporating additional features to improve security, functionality, and user convenience.

Possible future improvements include:

- Implementation of password storage using EEPROM or Flash memory to retain passwords even after power loss.
- Support for multiple users with separate passwords and access levels.
- Integration of LCD or OLED displays for improved user interaction.

- Addition of biometric authentication methods such as fingerprint recognition.
- Implementation of encrypted password storage for enhanced security.
- Integration with IoT platforms for remote monitoring and control.
- Development of a mobile application interface for password management.
- Addition of OTP (One-Time Password) based authentication for multi-factor security.
- Voice-based authentication and control.
- RFID card integration for access control.
- Real-Time Clock (RTC) integration for time-based access control.
- Event logging to record successful and failed login attempts.
- Admin mode for managing users and passwords.
- Password complexity checking to enforce stronger passwords.

8.3 Final Remarks

- The UART-Based Password Security System demonstrates how FPGA technology can be used to implement secure and reliable digital authentication systems efficiently. The project combines Finite State Machine (FSM) design, UART communication, password verification, counters, timers, and access control mechanisms into a single hardware implementation.
- The successful implementation confirms that FPGA-based solutions provide fast, reliable, and real-time operation for security applications. The system effectively performs password authentication, failed-attempt monitoring, lock control, and password update operations while maintaining stable performance.
- The project also provided valuable practical experience in Verilog HDL design, simulation, debugging, hardware testing, and FPGA implementation techniques.
- The developed system can serve as a foundation for more advanced security and access control systems incorporating features such as multiple-user authentication, biometric verification, IoT connectivity, and encrypted password management in future applications.

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